Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Core Materials Technology for Lithium-Metal Anode Stabilization Using Polymeric Interface Coatings			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification 1)	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical & Electronics		Electrical & Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	Battery Materials	High polymer materials	Material technology for energy and environmental industry purpose
Concept of Proposed Technology	The proposed technology is a core polymeric lithium-anode interface coating platform designed to stabilize lithium (Li) metal anodes across multiple battery formats, with primary relevance to secondary batteries and validation under chemically aggressive benchmark conditions. The central concept is to create thin, conformal, single Li^+-conducting polymer coatings that form an artificial, ion-selective interphase on Li metal, replacing or augmenting the native solid–electrolyte interphase (SEI) with a chemically robust, mechanically compliant, and electrochemically well-defined interface layer. The coating platform is based on chemically resistant polymer chemistries, including biphenyl-based single-ion conductors (BPSA) and polyolefin-derived cation-exchange polymers. These materials are engineered to deliver high Li^+ conductivity, high Li^+ transference number, low solvent permeability, and low swelling, while maintaining strong adhesion to lithium metal and long-term interfacial stability. By decoupling lithium-ion transport from solvent motion and suppressing parasitic interfacial reactions, the coatings provide a controlled and durable pathway for Li^+ transport across the anode–electrolyte interface.			

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Rationale for Joint R&D Project	The stabilization of Li metal anodes is a critical bottleneck for next-generation Li-metal batteries, limiting energy density, safety, and lifetime. This project addresses this challenge through a core polymeric Li-anode interface coating technology based on single Li^+-conducting, chemically robust polymers. The U.S. research institute provides advanced polymer and interface science expertise, while the Korean SME contributes industrial cell integration and commercialization capability. This complementary partnership enables parallel core technology development and technology internalization, accelerating scale-up, reducing technical risk, and aligning with TIPA's priorities in strategic materials, SME competitiveness, and industrial readiness.
Key Words	1. Polymeric Li-Anode Interface Coatings 2. Single-Ion Conducting Polymer Electrolytes (Single Li^+ Conductors) 3. Interface Stabilization Technology 4. Scalable Thin-Film Polymer Coating Platform 5. Cross-Platform Li-Metal Protection Technology

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)

R&D Objectives

This R&D project aims to develop advanced polymeric protective coatings for lithium (Li) metal anodes as a core enabling materials technology for next-generation Li-based batteries, with primary relevance to secondary battery systems and broad applicability across multiple battery formats. The central objective is to establish chemically robust, lithium ion (Li^+)–conductive polymer coatings that stabilize the Li/electrolyte interface, suppress parasitic interfacial reactions, and reduce impedance growth, thereby enabling safer, more reliable, and longer-lived lithium metal batteries.

Specifically, the project will design and synthesize chemically resistant biphenyl-based (BPSA) and polyolefin-based single Li^+-conducting polymers that exhibit high Li^+ selectivity, low solvent/redox agent permeability, and strong interfacial stability against diverse electrolytes relevant to secondary batteries [1]. Polymer architecture (e.g., ionic group density, crosslinking, and backbone chemistry) will be systematically optimized to achieve high Li^+ transport, low swelling, minimal interfacial resistance, and long-term dimensional and electrochemical stability under practical operating and storage conditions [2].

A key technical objective is to engineer thin, conformal, defect-controlled anode coating structures that provide effective physical and electrochemical barriers while maintaining fast Li^+ transport. Through precise control of coating thickness, morphology, and interfacial chemistry, the project aims to suppress dendrite-inducing side reactions, mitigate electrolyte decomposition, and stabilize the solid–electrolyte interface (SEI), thereby extending cycle life and improving safety in Li metal secondary batteries (**Figure 1**).

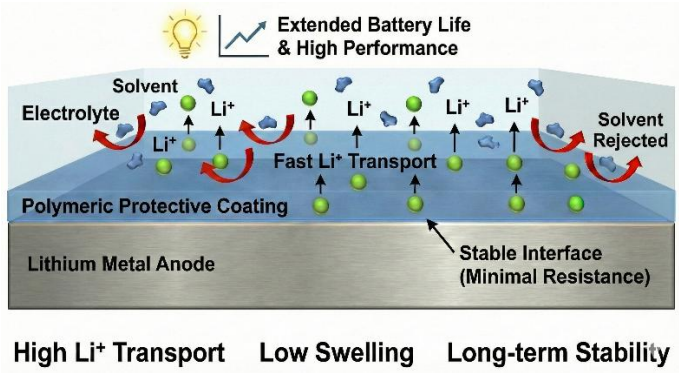

Figure 1. Operational mechanism and key benefits of the advanced polymeric protective coating on a Li metal anode. The schematic illustrates the coating's crucial role in enabling high-performance batteries. It demonstrates selective permeability, facilitating fast Li^+ transport from the anode to the electrolyte while effectively rejecting solvent molecules to suppress parasitic reactions. This protective barrier establishes a stable, low-resistance interface, minimizes swelling, and ensures long-term electrochemical stability, ultimately leading to extended battery life and enhanced overall performance.

To validate robustness under extreme conditions, the developed polymer coatings will be applied to Li–$SOCl_2$ batteries as a stringent demonstration platform, where highly corrosive electrolytes and severe interfacial chemistry provide an accelerated test environment. In this context, the project will demonstrate controlled LiCl passivation behavior, reduced interfacial impedance growth, and significant reduction of voltage delay after long-term storage, thereby proving the chemical resilience and interfacial control capability of the protective coating.

At the system level, the project will integrate the developed coatings into prototype lithium battery cells, with primary relevance to secondary battery formats and Li–$SOCl_2$ cells used as a representative extreme-condition case study. In parallel, scalable coating and manufacturing processes compatible with existing anode fabrication lines will be established to enable rapid technology transfer, technology internalization, and commercialization by Korean SMEs.

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R&D Target Specifications

The R&D target specifications are defined to ensure that the developed polymeric Li-anode protective coatings deliver high Li^+ transport, controlled interfacial chemistry, and long-term chemical and mechanical durability across a broad range of Li-based battery systems and electrolyte environments, including both secondary batteries and chemically aggressive benchmark systems used for accelerated validation.

To validate robustness under extreme chemical and interfacial conditions, $LiAlCl_4/SOCl_2$ electrolytes are used as a stringent benchmark system, providing an accelerated test platform for chemical stability, interfacial control, and passivation behavior. Based on this dual-use framework (relevance with extreme-condition validation), we have established the following quantitative performance targets, benchmarked against current commercial Li-anode and coating technologies.

Category	Key Performance Indicator (KPI)	Current Standard (Uncoated)	Project Final Target	Test / Measurement Condition
Li^+ Transport	Ionic Conductivity, S/cm	Not defined (Li^+ transport via native LiCl passivation layer)	$>10^{-4}$	@ 25-40◦C
	Li^+ Transference Number (t^+)	<0.5 (Liquid electrolyte)	≥0.9	BPSA-based single-ion conductor
Protective Function	$SOCl_2$ Permeability, %	- Initial direct $SOCl_2$ access to bare Li: ~100% (before LiCl forms). - Steady-state $SOCl_2$ permeability in LiCl layer: Very low	≤5%	- Solvent uptake - $SOCl_2$ permeability measurement - Electrochemical access metrics (EIS/ASR)
Stability	Chemical Stability (Mass Loss), wt %	N/A (Li corrodes)	≤1.0%	Immersion in $LiAlCl_4/SOCl_2$ (1,000h @ 60 °C)
	Swelling Ratio, %	N/A	≤5.0%	Volume increase after immersion
Interfacial Control	Total Interfacial Resistance, $\Omega\cdot cm^2$	>50 (After storage)	≤40-50	EIS/ASR
Cell Performance	Voltage Delay ($t_{3.0V}$), sec	10 - 100	≤5-10	Time to reach 3.0 V under standard load (25 °C/12mo equivalent)
	Capacity Retention, %	90~95	≥90%	After 1 year @ 25◦C
Reliability	Thermal Stability	$Li/LiCl/SOCl_2$ interface remains functional but interfacial resistance increases <100 h	≥ 500 h with no coating delamination, cracking, or catastrophic impedance rise	Coated Li in $LiAlCl_4/SOCl_2$ held at 60–70 °C; periodic EIS and visual inspection

R&D Plan

The project follows a materials-first, interface-driven development pathway to establish polymeric lithium-anode protective coatings as a core, cross-platform interface technology for lithium-based batteries. The research will proceed through:

(1) the design and synthesis of chemically robust, single Li^+-conducting polymers (BPSA and polyolefin-based CEM variants) with relevance to diverse Li battery electrolytes;

(2) quantitative evaluation of physicochemical and electrochemical properties across representative electrolyte environments, including chemically aggressive benchmark systems;

(3) engineering of thin, defect-controlled coatings that stabilize the Li/electrolyte interface and regulate interfacial passivation behavior;

(4) performance validation in prototype Li battery cells, using $Li–SOCl_2$ as a stringent extreme-condition demonstration platform with emphasis on voltage delay, interfacial impedance, and storage stability; and

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(5) development of manufacturing-ready coating protocols and quality-control guidelines to enable technology internalization and commercialization by Korean SMEs.

Timeline Summary (3-Year)

- **Year 1:** Polymer synthesis + stability/transport screening; initial coating feasibility
- **Year 2:** Coating optimization + LiCl morphology control; early prototype cell validation
- **Year 3:** Extended storage/voltage delay validation; tech transfer + manufacturing/QC readiness for SME commercialization

Task	Methods	KPI Targets	Deliverables
Task 1. Polymer Design & Synthesis	Synthesize BPSA and polyolefin-based CEM polymers; tune ionic group density, crosslinking, backbone chemistry	Stable polymer films; no dissolution in $LiAlCl_4/SOCl_2$; swelling ≤5%	Polymer library; structure–property dataset
Task 2. Physicochemical Stability Screening	Immersion tests (25–60 °C), FTIR/NMR, DSC/TGA, SEM, mass change	Mass loss ≤1 wt%; no cracking/delamination after 1,000 h	Stability-qualified polymer shortlist
Task 3. Electrochemical Screening-Materials Level	EIS, t+ measurement, oxidative stability tests, impedance	Li+ conductivity ≥10^{-4} S cm-1; t+ ≥0.9; impedance ≤40-50 Ω·cm^2	Electrochemical performance report
Task 4. Thin-Film Coating Engineering	Solution/spray coating, lamination, casting.	Thickness < 5 μm; ≥99% coverage; defect density ≤1 cm^{-2}	Defect-controlled coated Li foils
Task 5. Adhesion & Mechanical Integrity Testing	Peel tests, thermal cycling, bending tests, electrolyte soaking	No delamination; no cracking after 500 h at 60 °C	Adhesion-qualified coatings
Task 6. LiCl Passivation Control	Interface tuning, EIS deconvolution, SEM/EDS, XPS	Porous/fine LiCl; no compact crust >5–10 μm; ΔR ≤50 Ω·cm2	Controlled LiCl interface dataset
Task 7. Prototype Cell Integration	Assemble coin/cylindrical Li–$SOCl_2$ cells with coated anodes	Activation time ≤10 s; overshoot ≤200 mV	Prototype cell performance report
Task 8. Voltage Delay & Storage Validation	Long-term storage (25 °C), accelerated aging (60–70 °C), EIS	Voltage delay ≤5–10 s; impedance growth ≤2×	Storage-life validation report
Task 9. Safety & Reliability Testing	Gas evolution indicators, swelling, voltage stability	No abnormal swelling/gassing; stable discharge	Safety assessment dataset
Task 10. Manufacturing Readiness & Tech Transfer	Process window definition; QC metrics; SOP creation	Thickness ±10%; impedance ±15% batch-to-batch	SOP + QC + tech-transfer package
Task 11. Commercialization Preparation	Pilot planning, cost drivers, SME integration roadmap	Pilot readiness; defined throughput targets	Commercialization roadmap

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**A. Technical Leadership and Core Materials Innovation** RPI will serve as the technical lead for advancing polymeric Li-anode protective coating technologies as a core enabling materials platform for Li-metal batteries across multiple formats, with relevance to secondary batteries and validation under extreme electrolyte environments. RPI will lead the design, synthesis, and optimization of chemically robust single Li^+-conducting polymers, including BPSA systems and chemically resistant polyolefin-based cation-exchange polymers. A central responsibility will be to establish structure–property–performance relationships linking polymer backbone chemistry, ionic group density, crosslinking degree, and interfacial chemistry to Li^+ transport, solvent permeability, interfacial stability, passivation behavior, and interfacial impedance. These insights will guide the rational design of polymer coatings capable of stabilizing the Li/electrolyte interface across diverse electrolyte chemistries. **B. Research Scope** Within the defined research scope, RPI will conduct polymer design and synthesis, including systematic variation of polymer architectures to achieve high Li^+ conductivity/transference number, low swelling, and long-term chemical stability in representative lithium battery electrolytes. Chemically aggressive catholyte systems, including $LiAlCl_4/SOCl_2$, will be used as benchmark environments to accelerate validation of chemical and interfacial robustness. RPI will also perform comprehensive physicochemical characterization, including thermal stability (DSC/TGA), chemical integrity (FTIR/NMR), morphology (SEM), and swelling and mass-change analysis under corrosive catholyte exposure. In parallel, RPI/UIC will lead electrochemical characterization at the materials and interface levels, including ionic conductivity (EIS), Li^+ transference number (t^+), oxidative stability, solvent permeability, and Li/polymer interfacial impedance evolution. These measurements will provide quantitative benchmarks for evaluating coating performance against defined KPI targets. **C. Interface Engineering and Passivation Control** A key technical responsibility of RPI is the engineering of Li-anode coating structures that stabilize the Li/electrolyte interface while suppressing parasitic reactions and excessive impedance growth. RPI will develop thin, conformal, defect-controlled polymer coatings on lithium foil and systematically tune coating thickness, morphology, and interfacial chemistry to regulate interfacial passivation behavior. As part of extreme-condition validation, RPI will investigate controlled LiCl passivation at the polymer/$SOCl_2$ interface to establish design principles for regulated salt-film formation, morphology control, and impedance minimization. Post-mortem and in situ diagnostics (e.g., EIS deconvolution, SEM/EDS, XPS where applicable) will be used to correlate passivation-layer morphology with interfacial resistance, voltage-delay behavior, and long-term interface stability. **D. Resources and Infrastructure** RPI will provide access to specialized polymer synthesis laboratories, electrochemical testing facilities, and analytical instrumentation required to execute the project. Dedicated research personnel with expertise in polymer chemistry, electrochemistry, and battery interfaces will be assigned to the project. Laboratory-scale coating and film-formation capabilities will be maintained to support thin-film coating development and interface engineering studies. **E. Knowledge Generation, Transfer, and Collaboration** RPI will play a central role in knowledge generation and technology transfer to the Korean SME partner. This includes sharing polymer formulations, coating protocols, characterization methods, and performance data to support downstream cell integration and commercialization. RPI will also participate in regular technical meetings, joint data reviews, milestone assessments, and collaborative problem-solving, and will contribute to technical reporting and dissemination of non-confidential results in accordance with project agreements.

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Expected R&R of Korean SME Partner	**A. Prototype and Commercial-Format Cell Assembly** A core responsibility of the Korean SME will be the assembly of laboratory-to-commercially relevant Li battery cells incorporating polymer-coated Li anodes. This includes fabrication of prototype cell formats (e.g., coin, cylindrical, or application-specific formats as appropriate), optimization of anode handling and coating integration procedures, and refinement of stack pressure, separator compatibility, and electrolyte filling protocols. While Li–$SOCl_2$ cells will be used as a stringent benchmark and extreme-condition validation platform, the assembly activities will be structured to ensure relevance to broader Li-metal battery formats, including secondary batteries. Emphasis will be placed on manufacturability, mechanical robustness, reproducibility, and consistency with existing production workflows. **B. Commercial-Level Cell Integration and Product Development** The Korean SME partner will serve as the industrial implementation and commercialization lead, with primary responsibility for translating polymeric Li-anode protective coatings developed by the research institute into commercially relevant lithium battery products. The SME will focus on cell design, anode integration, and product engineering to ensure that the developed polymer coatings are compatible with practical anode fabrication, electrode handling, cell assembly, and packaging processes used in industrial lithium battery manufacturing. **C. Testing, Validation, and Performance Qualification** The SME partner will lead cell-level performance validation and qualification activities using testing protocols aligned with commercial Li-metal battery evaluation. The SME will also conduct reliability and preliminary safety assessments, including checks for abnormal gas evolution, swelling, voltage instability, and internal shorting behavior. These data will be used to verify that the coated-anode technology meets defined KPI targets and product qualification requirements. **D. Commercialization Preparation and Manufacturing Readiness** The Korean SME will play a central role in commercialization planning and manufacturing readiness assessment. This includes evaluation of process compatibility with existing Li anode fabrication and cell assembly lines, definition of preliminary quality-control (QC) acceptance criteria (e.g., coating integrity, voltage-delay thresholds, impedance limits), and identification of cost drivers and supply-chain considerations. The SME will develop a commercialization roadmap outlining next steps for pilot production, certification, and market entry of polymer-coated lithium-anode technologies, with initial deployment in Li–$SOCl_2$ products and subsequent expansion to secondary-battery platforms. **E. Collaboration, Technology Internalization, and Feedback to R&D** The Korean SME will work in close collaboration with RPI through regular technical coordination meetings, data exchange, and joint troubleshooting. A key responsibility will be technology internalization, ensuring that coating integration know-how, performance criteria, and product qualification standards are embedded within domestic manufacturing capabilities. Feedback from cell testing and commercialization-oriented evaluations will be systematically communicated to the research institute to guide further materials optimization and interface engineering.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint R&D project is expected to deliver a core polymeric lithium-anode interface technology that improves the reliability, safety, and lifetime of lithium-metal batteries across multiple formats, with direct relevance to secondary battery systems. The primary technological outcome will be the development of chemically robust, single Li^+-conducting polymer coatings exhibiting high Li^+ conductivity, high Li^+ transference number, low swelling, and long-term chemical and electrochemical stability across diverse electrolyte environments. A key technical achievement will be the demonstration of thin, conformal, defect-controlled coatings that stabilize the Li/electrolyte interface while suppressing parasitic reactions and interfacial impedance growth. Through precise control of coating thickness, morphology, and interfacial chemistry, the developed coatings are expected to mitigate dendrite-inducing side reactions, stabilize the solid–electrolyte interphase (SEI), and enable durable lithium-metal operation in secondary batteries. To validate robustness under extreme chemical and interfacial conditions, the polymer coatings will be applied to Li–$SOCl_2$ batteries as a stringent benchmark platform. In this context, the project will demonstrate controlled LiCl passivation at the polymer/$SOCl_2$ interface and regulation of LiCl morphology toward porous or finely dispersed structures, significantly reducing interfacial impedance growth and voltage delay after long-term storage. These results will provide accelerated proof of chemical resilience and interface-control capability. At the system level, the project will demonstrate prototype lithium battery cells incorporating polymer-coated lithium anodes, with primary relevance to secondary-battery formats and Li–$SOCl_2$ used as an extreme-condition case study. These cells are expected to exhibit improved voltage activation behavior, reduced voltage overshoot, enhanced discharge efficiency, improved storage stability, and superior interfacial durability relative to uncoated anodes. From a commercialization perspective, the project is structured to enable technology internalization by the Korean SME partner. Expected outcomes include polymer-coating formulations compatible with existing Li-anode fabrication lines, defined QC acceptance criteria, and a commercialization roadmap for pilot production and product qualification. This will provide the SME with a differentiated, proprietary anode-protection technology deployable initially in Li–$SOCl_2$ products and subsequently expandable to secondary-battery and Li-metal battery markets. Overall, this project will deliver novel materials technology, commercialization-ready battery prototypes, and long-term industrial value, aligning with national priorities in advanced batteries, energy security, SME competitiveness, and sustainable industrial growth.

6. References

1. Wang, T., et al., *High-efficient multifunctional electrochemical membrane for lithium polysulfide redox flow batteries.* Journal of Membrane Science, 2021. **636**: p. 119539.
2. Zhou, H., et al., *Protective coatings for lithium metal anodes: Recent progress and future perspectives.* Journal of Power Sources, 2020. **450**: p. 227632.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization